BRIEF INSIGHTS + QUERY RESULTS OF SQL SCRIPT

1. **Load Dataset into SQL Database** - This step creates the table structure and prepares the database for storing sales records.

   Output-

Data Output Messages Notifications

Showing rows: 9001 to 9994 Page No: 10 of 10

	row_id integer	order_id character varying (50)	order_date character varying (20)	ship_date character varying (20)	ship_mode character varying (50)	customer_id character varying (50)	customer_name character varying (100)	segment character varying
9001	9001	CA-2014-133389	6/22/2014	6/22/2014	Same Day	TB-21280	Toby Braunhardt	Consumer
9002	9002	CA-2014-103807	12/2/2014	12/9/2014	Standard Class	EM-13825	Elizabeth Moffitt	Corporate
9003	9003	CA-2014-103807	12/2/2014	12/9/2014	Standard Class	EM-13825	Elizabeth Moffitt	Corporate
9004	9004	CA-2015-102855	4/5/2015	4/6/2015	Same Day	JF-15415	Jennifer Ferguson	Consumer
9005	9005	CA-2015-102855	4/5/2015	4/6/2015	Same Day	JF-15415	Jennifer Ferguson	Consumer
9006	9006	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9007	9007	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9008	9008	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9009	9009	CA-2017-107825	11/18/2017	11/18/2017	Same Day	NB-18655	Nona Balk	Corporate
9010	9010	CA-2017-106047	7/12/2017	7/18/2017	Standard Class	CS-11860	Cari Schnelling	Consumer
9011	9011	CA-2017-127803	5/19/2017	5/23/2017	Standard Class	ND-18460	Neil Ducich	Corporate
9012	9012	CA-2015-137071	12/20/2015	12/21/2015	First Class	ED-13885	Emily Ducich	Home Office
9013	9013	CA-2016-121447	2/21/2016	2/22/2016	First Class	EA-14035	Erin Ashbrook	Corporate
9014	9014	CA-2017-143084	3/18/2017	3/22/2017	Standard Class	BP-11050	Barry Pond	Corporate
9015	9015	CA-2017-143084	3/18/2017	3/22/2017	Standard Class	BP-11050	Barry Pond	Corporate
9016	9016	CA-2016-131639	12/17/2016	12/23/2016	Standard Class	NS-18640	Noel Staavos	Corporate
9017	9017	CA-2016-131639	12/17/2016	12/23/2016	Standard Class	NS-18640	Noel Staavos	Corporate

Total rows: 9994 Query complete 00:00:00.275 CRLF Ln 25, Col 15

2. **Create Dimensional tables**- In this particular step we create tables (customers, orders, products) from the dataset and Insert data into these tables using SELECT DISTINCT.

1. **Customers Table**- This table stores unique customer information.

Output-

SELECT *FROM customers LIMIT 10;

Showing rows: 1 to 10 Page No: 1 of 1

	customer_id [PK] character varying (20)	customer_name character varying (100)	segment character varying (50)	country character varying (50)	city character varying (100)	state character varying (100)	postal_code character varying (20)	region character varying (50)
1	AA-10315	Alex Avila	Consumer	United States	Minneapolis	Minnesota	55407	Central
2	AA-10375	Allen Armold	Consumer	United States	Lebanon	Tennessee	37087	South
3	AA-10480	Andrew Allen	Consumer	United States	Middletown	Connecticut	6457	East
4	AA-10645	Anna Andreadi	Consumer	United States	Georgetown	Kentucky	40324	South
5	AB-10015	Aaron Bergman	Consumer	United States	Arlington	Texas	76017	Central
6	AB-10060	Adam Bellavance	Home Office	United States	Greenwood	Indiana	46142	Central
7	AB-10105	Adrian Barton	Consumer	United States	Phoenix	Arizona	85023	West
8	AB-10150	Aimee Bixby	Consumer	United States	Yonkers	New York	10701	East
9	AB-10165	Alan Barnes	Consumer	United States	Toledo	Ohio	43615	East
10	AB-10255	Alejandro Ballentine	Home Office	United States	Jackson	Mississippi	39212	South

Total rows: 10 Query complete 00:00:00.244 CRLF Ln 144, Col 10

2. **<u>Produscts Table-</u>** This table stores unique product details.

<u>Output</u>- SELECT * FROM products LIMIT 10;

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	product_id [PK] character varying (50)	product_name character varying (255)	category character varying (50)	sub_category character varying (50)
1	FUR-BO-10000112	Bush Birmingham Collection Bookcase, Dark Cherry	Furniture	Bookcases
2	FUR-BO-10000330	Sauder Camden County Barrister Bookcase, Planked Cherry Finish	Furniture	Bookcases
3	FUR-BO-10000362	Sauder Inglewood Library Bookcases	Furniture	Bookcases
4	FUR-BO-10000468	O'Sullivan 2-Shelf Heavy-Duty Bookcases	Furniture	Bookcases
5	FUR-BO-10000711	Hon Metal Bookcases, Gray	Furniture	Bookcases
6	FUR-BO-10000780	O'Sullivan Plantations 2-Door Library in Landvery Oak	Furniture	Bookcases
7	FUR-BO-10001337	O'Sullivan Living Dimensions 2-Shelf Bookcases	Furniture	Bookcases
8	FUR-BO-10001519	O'Sullivan 3-Shelf Heavy-Duty Bookcases	Furniture	Bookcases
9	FUR-BO-10001567	Bush Westfield Collection Bookcases, Dark Cherry Finish, Fully Assem...	Furniture	Bookcases
10	FUR-BO-10001601	Sauder Mission Library with Doors, Fruitwood Finish	Furniture	Bookcases

Total rows: 10 Query complete 00:00:00.158 CRLF Ln 144, Col 24

3. **<u>Orders Table-</u>** This table stores the Transactional sales and connect customers and products together.

<u>Output</u>- SELECT * FROM orders LIMIT 10;

Data Output Messages Notifications

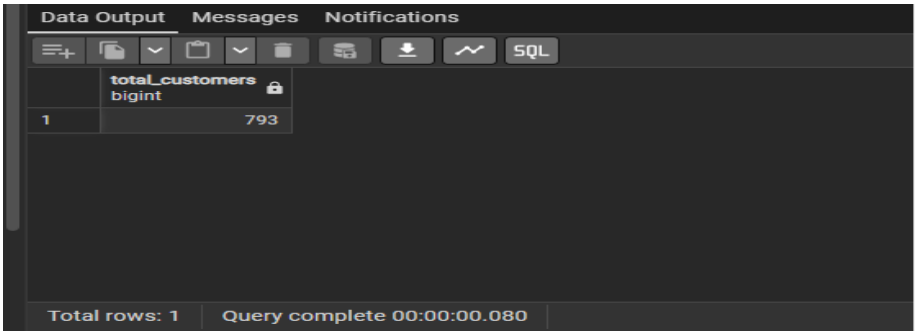
Showing rows: 1 to 10 Page No: 1 of 1

	row_id [PK] integer	order_id character varying (50)	order_date date	ship_date date	ship_mode character varying (50)	customer_id character varying (20)	product_id character varying (50)	sales numeric (10,2)	quantity integer	discount numeric (5,2)	profit numeric (10,2)
1	8020	CA-2017-167227	2017-11-02	2017-11-05	First Class	NP-18670	OFF-AP-10001962	83.90	10	0.00	20.98
2	5033	CA-2016-155166	2016-12-26	2017-01-02	Standard Class	BB-11545	FUR-CH-10003968	212.94	3	0.00	25.55
3	9136	CA-2016-124254	2016-04-12	2016-04-17	Standard Class	TB-21175	OFF-ST-10000060	51.98	1	0.20	-5.20
4	4914	CA-2017-127306	2017-01-14	2017-01-18	Standard Class	BH-11710	OFF-BI-10003727	2.51	2	0.70	-1.84
5	5648	CA-2017-168179	2017-11-02	2017-11-07	Standard Class	JB-16000	OFF-PA-10001800	19.44	3	0.00	9.33
6	1406	US-2014-118486	2014-04-06	2014-04-08	First Class	SD-20485	FUR-TA-10001039	154.76	3	0.40	-36.11
7	9116	CA-2017-153045	2017-01-19	2017-01-24	Second Class	NS-18505	TEC-PH-10001817	429.60	2	0.40	-93.08
8	7156	US-2015-136749	2015-12-25	2015-12-27	Second Class	LH-16900	OFF-BI-10004492	157.90	5	0.00	74.21
9	5135	CA-2017-155607	2017-11-26	2017-11-30	Standard Class	FG-14260	OFF-BI-10003364	93.34	4	0.20	32.67
10	8250	CA-2015-140221	2015-03-05	2015-03-09	Second Class	MS-17365	OFF-BI-10002854	11.21	2	0.80	-16.82

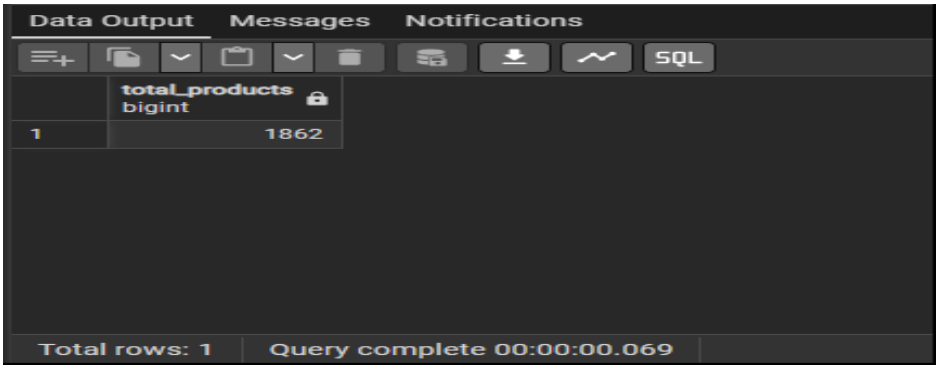
Total rows: 10 Query complete 00:00:00.121 CRLF Ln 146, Col CRLF Ln 146, Col 23

4. **Number Of Records:-**

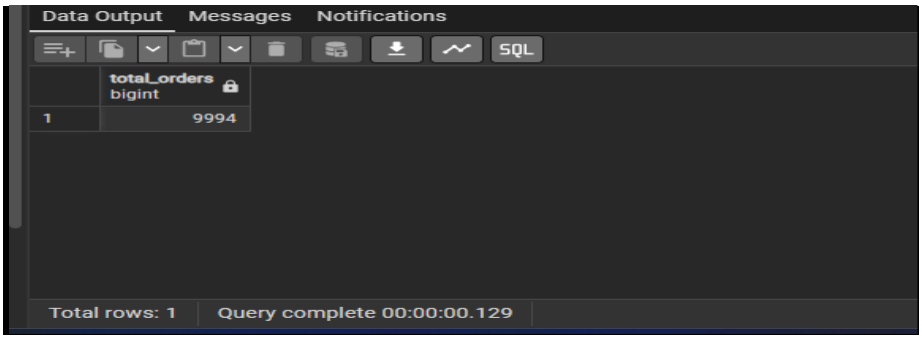
Customers



Products



Orders



3. **Apply Subqueries**–

1. **Customers with Above Average Sales** -This Query gives orders with sales greater than overall average sales.

Output-

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 3

	row_id [PK] integer	order_id character varying (50)	order_date date	ship_date date	ship_mode character varying (50)	customer_id character varying (20)	product_id character varying (50)	sales numeric (10,2)	quantity integer	dis nur
1	9116	CA-2017-153045	2017-01-19	2017-01-24	Second Class	NS-18505	TEC-PH-10001817	429.60	2	
2	8610	CA-2014-151946	2014-06-04	2014-06-09	Standard Class	BT-11440	FUR-BO-10003272	353.57	2	
3	6877	US-2016-121013	2016-09-05	2016-09-10	Standard Class	MM-17920	FUR-TA-10003238	1652.94	3	
4	2404	US-2016-110170	2016-09-27	2016-10-03	Standard Class	HM-14860	FUR-BO-10000780	956.66	7	
5	407	CA-2017-117457	2017-12-08	2017-12-12	Standard Class	KH-16510	TEC-CO-10004115	1199.98	3	
6	3769	CA-2014-153913	2014-12-16	2014-12-20	Second Class	KB-16585	FUR-CH-10000988	1013.83	9	
7	5038	CA-2017-141719	2017-11-17	2017-11-21	Second Class	EG-13900	TEC-AC-10003610	239.96	5	
8	5321	US-2017-162558	2017-10-02	2017-10-05	First Class	Dp-13240	FUR-TA-10000198	2314.12	7	
9	6952	CA-2015-161767	2015-11-20	2015-11-24	Standard Class	GK-14620	TEC-MA-10002790	479.99	2	
10	2494	CA-2016-146206	2016-09-10	2016-09-14	Second Class	KT-16480	FUR-TA-10004086	300.93	5	
11	1305	US-2017-126179	2017-07-03	2017-07-07	Standard Class	CS-12460	TEC-PH-10001557	287.97	3	
12	9593	US-2016-105452	2016-07-28	2016-08-01	Standard Class	BF-11005	FUR-FU-10003806	302.72	5	
13	6115	CA-2016-124590	2016-11-12	2016-11-16	Standard Class	SP-20920	FUR-CH-10003746	449.37	2	
14	4867	CA-2017-122490	2017-11-13	2017-11-18	Standard Class	TT-21070	OFF-BI-10001718	563.02	11	
15	1631	CA-2016-106530	2016-05-08	2016-05-08	Same Day	CL-12565	OFF-ST-10002011	1006.06	3	
16	5211	CA-2015-156755	2015-01-12	2015-01-18	Standard Class	YS-21880	OFF-ST-10002615	465.18	3	
17	2685	CA-2017-137085	2017-06-29	2017-07-06	Standard Class	CT-11995	OFF-BI-10000632	312.55	9	

Total rows: 2360 Query complete 00:00:00.275 CRLF Ln 166, Col 3

2. **Highest Order Per Customer** - This Returns highest-value order of every customer.
   Output-

Data Output Messages Notifications

	customer_id character varying (20)	order_id character varying (50)	sales numeric (10,2)
1	BT-11440	CA-2014-151946	353.57
2	MM-17920	US-2016-121013	1652.94
3	KB-16585	CA-2014-153913	1013.83
4	CS-12460	US-2017-126179	287.97
5	RE-19405	CA-2014-165477	48.36
6	DK-12985	CA-2014-163034	646.20
7	SM-20005	CA-2014-148488	106.34
8	BP-11290	CA-2016-142398	638.29
9	JC-15775	CA-2016-118626	705.54
10	AB-10600	US-2017-155425	899.14
11	JH-16180	CA-2015-138898	845.73
12	LP-17095	CA-2017-121790	2003.17
13	CS-11950	US-2014-141257	585.55
14	AS-10240	CA-2014-149958	142.78
15	TS-21340	CA-2015-153416	454.86
16	SM-20905	CA-2016-129280	431.98
17	BT-11485	CA-2015-165085	355.96

Total rows: 795 Query complete 00:00:28.597

4. **Apply CTE Function-**

1. **Total Sales Per Customer –** Shows total sales generated by each customer.

Output-

Data Output Messages Notifications

	customer_id character varying (20)	total_sales numeric
1	SM-20320	25043.07
2	TC-20980	19052.22
3	RB-19360	15117.35
4	TA-21385	14595.62
5	AB-10105	14473.57
6	KL-16645	14175.23
7	SC-20095	14142.34
8	HL-15040	12873.30
9	SE-20110	12209.44
10	CC-12370	12129.08
11	TS-21370	11891.75
12	GT-14710	11820.12
13	BM-11140	11789.64
14	SV-20365	11470.94
15	CJ-12010	11164.97
16	CL-12565	10880.56
17	ME-17320	10663.73

Total rows: 793 Query complete 00:00:00.207

2. **Total Profit Per Customer-** Shows total profit generated by each customer.

Output-

Data Output Messages Notifications

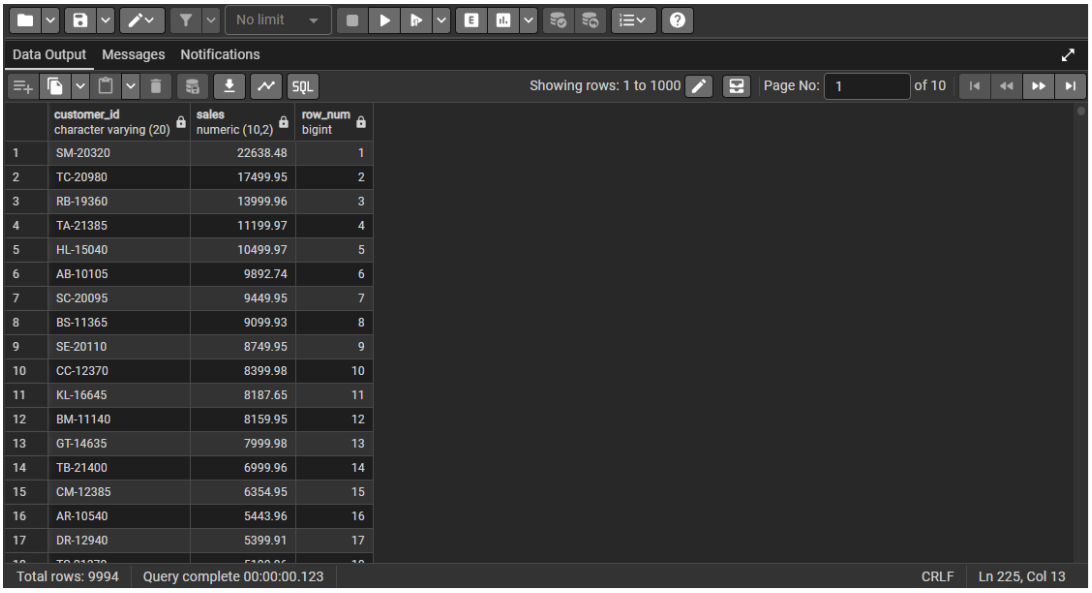
	customer_id character varying (20)	total_profit numeric
1	TC-20980	8981.32
2	RB-19360	6976.09
3	SC-20095	5757.42
4	HL-15040	5622.43
5	AB-10105	5444.81
6	TA-21385	4703.80
7	CM-12385	3899.91
8	KD-16495	3038.58
9	AR-10540	2884.61
10	DR-12940	2869.08
11	TB-21400	2798.37
12	NM-18445	2751.69
13	SE-20110	2650.67
14	BS-11365	2616.06
15	HM-14860	2437.97
16	TS-21370	2371.73
17	BM-11650	2199.28

Total rows: 793 Query complete 00:00:00.240

5. **Window Functions** -They are the functions that performs calculations across a set of rows related to the current row without grouping the rows into a single result.

1. **ROW_NUMBER():** Assigns a unique sequential number to each row within a result set.

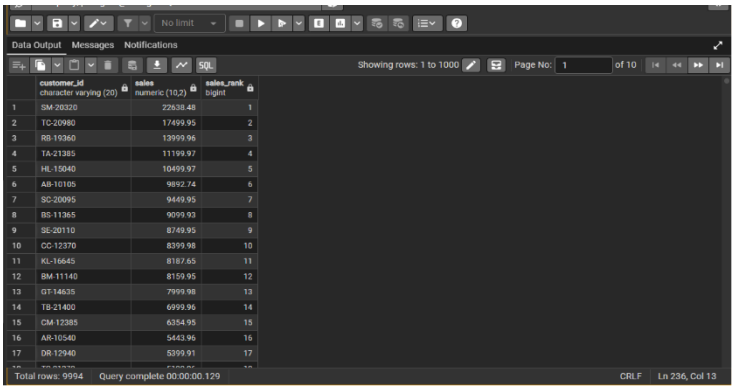
Output-



	customer_id character varying (20)	sales numeric (10,2)	row_num bigint
1	SM-20320	22638.48	1
2	TC-20980	17499.95	2
3	RB-19360	13999.96	3
4	TA-21385	11199.97	4
5	HL-15040	10499.97	5
6	AB-10105	9892.74	6
7	SC-20095	9449.95	7
8	BS-11365	9099.93	8
9	SE-20110	8749.95	9
10	CC-12370	8399.98	10
11	KL-16645	8187.65	11
12	BM-11140	8159.95	12
13	GT-14635	7999.98	13
14	TB-21400	6999.96	14
15	CM-12385	6354.95	15
16	AR-10540	5443.96	16
17	DR-12940	5399.91	17

2. **RANK():** Assigns ranks to rows, giving the same rank to ties and skipping subsequent rank numbers.

Output-



	customer_id character varying (20)	sales numeric (10,2)	sales_rank bigint
1	SM-20320	22638.48	1
2	TC-20980	17499.95	2
3	RB-19360	13999.96	3
4	TA-21385	11199.97	4
5	HL-15040	10499.97	5
6	AB-10105	9892.74	6
7	SC-20095	9449.95	7
8	BS-11365	9099.93	8
9	SE-20110	8749.95	9
10	CC-12370	8399.98	10
11	KL-16645	8187.65	11
12	BM-11140	8159.95	12
13	GT-14635	7999.98	13
14	TB-21400	6999.96	14
15	CM-12385	6354.95	15
16	AR-10540	5443.96	16
17	DR-12940	5399.91	17

3. **DENSE_RANK():** Assigns ranks to rows, giving the same rank to ties but without skipping any rank numbers.

Output-

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 10

	customer_id character varying (20)	sales numeric (10,2)	dense_rank_no bigint
1	SM-20320	22638.48	1
2	TC-20980	17499.95	2
3	RB-19360	13999.96	3
4	TA-21385	11199.97	4
5	HL-15040	10499.97	5
6	AB-10105	9892.74	6
7	SC-20095	9449.95	7
8	BS-11365	9099.93	8
9	SE-20110	8749.95	9
10	CC-12370	8399.98	10
11	KL-16645	8187.65	11
12	BM-11140	8159.95	12
13	GT-14635	7999.98	13
14	TB-21400	6999.96	14
15	CM-12385	6354.95	15
16	AR-10540	5443.96	16
17	DR-12940	5399.91	17

Total rows: 9994 Query complete 00:00:00.226 CRLF Ln 248, Col 13

6. **Combining JOIN + CTE + Window Functions:-** This combined query calculates the total sales generated by each customer and ranks them based on their sales contribution. It helps identify the highest-value customers and compare customer performance. Customers with the same total sales receive the same rank. The result can be used for customer segmentation and targeted business strategies.

Output:-

Data Output Messages Notifications

Showing rows: 1 to 793 Page No: 1 of 1

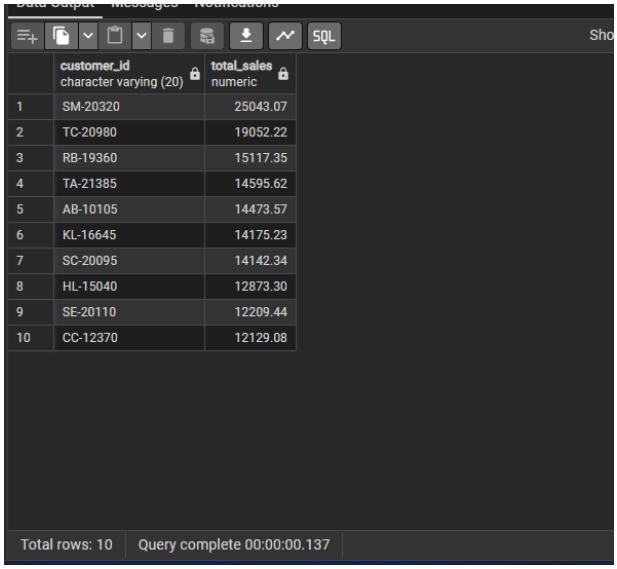
	customer_id [PK] character varying (20)	customer_name character varying (100)	total_sales numeric	customer_rank bigint
1	SM-20320	Sean Miller	25043.07	1
2	TC-20980	Tamara Chand	19052.22	2
3	RB-19360	Raymond Buch	15117.35	3
4	TA-21385	Tom Ashbrook	14595.62	4
5	AB-10105	Adrian Barton	14473.57	5
6	KL-16645	Ken Lonsdale	14175.23	6
7	SC-20095	Sanjit Chand	14142.34	7
8	HL-15040	Hunter Lopez	12873.30	8
9	SE-20110	Sanjit Engle	12209.44	9
10	CC-12370	Christopher Conant	12129.08	10
11	TS-21370	Todd Sumrall	11891.75	11
12	GT-14710	Greg Tran	11820.12	12
13	BM-11140	Becky Martin	11789.64	13
14	SV-20365	Seth Vernon	11470.94	14
15	CJ-12010	Caroline Jumper	11164.97	15
16	CL-12565	Clay Ludtke	10880.56	16
17	ME-17320	Maria Etezadi	10663.73	17

Total rows: 793 Query complete 00:00:00.231 CRLF Ln 272, Col 24

7. **Business queries:-** Displays top customers, low customers, single-order customers, above-average sales.

1. **Top-customers-** Display top 10 customers having highest sales.

Output-

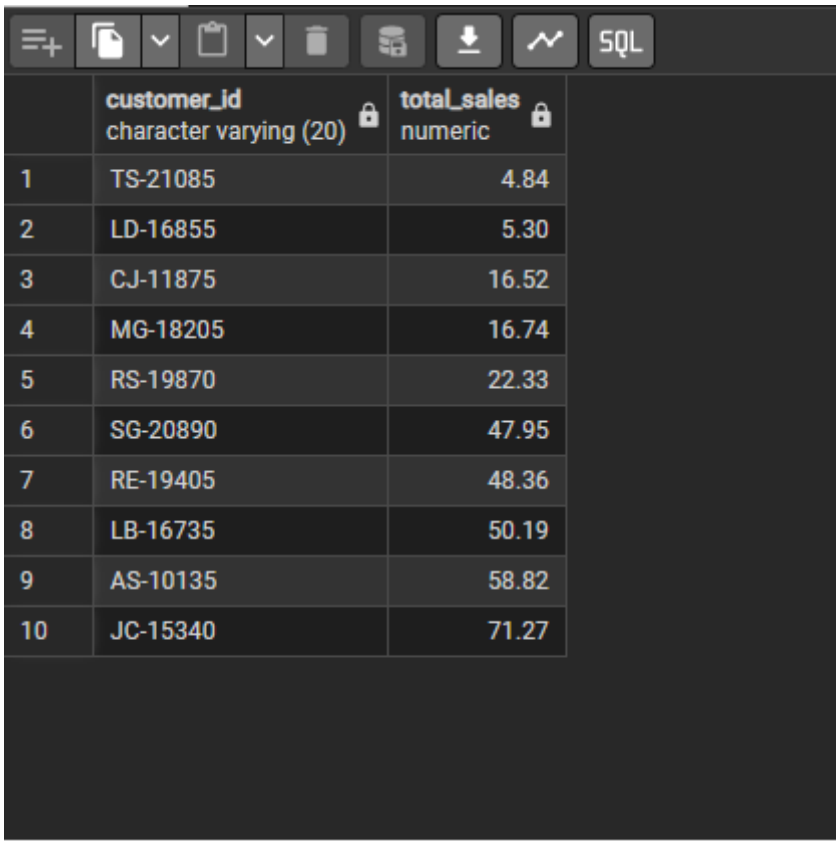


	customer_id character varying (20)	total_sales numeric
1	SM-20320	25043.07
2	TC-20980	19052.22
3	RB-19360	15117.35
4	TA-21385	14595.62
5	AB-10105	14473.57
6	KL-16645	14175.23
7	SC-20095	14142.34
8	HL-15040	12873.30
9	SE-20110	12209.44
10	CC-12370	12129.08

Total rows: 10 | Query complete 00:00:00.137

2. **Low Customers-** Display top 10 customers having lowest sales.

Output-



	customer_id character varying (20)	total_sales numeric
1	TS-21085	4.84
2	LD-16855	5.30
3	CJ-11875	16.52
4	MG-18205	16.74
5	RS-19870	22.33
6	SG-20890	47.95
7	RE-19405	48.36
8	LB-16735	50.19
9	AS-10135	58.82
10	JC-15340	71.27

3. **Single-Order Customers**- Customers who placed exactly one order.

Output-

Data Output Messages Notifications

	customer_id character varying (20)	total_orders bigint
1	AO-10810	1
2	AR-10570	1
3	CJ-11875	1
4	JC-15385	1
5	JR-15700	1
6	LD-16855	1
7	MG-18205	1
8	PH-18790	1
9	RE-19405	1
10	RM-19750	1
11	SM-20905	1
12	TC-21145	1

Total rows: 12 Query complete 00:00:00.136

4. **Customers Above Average Total Sales**- High-value customers contributing more than average revenue.

Output-

	customer_id character varying (20)	total_sales numeric
1	JP-15520	3635.59
2	JD-16150	8828.04
3	RF-19345	3309.14
4	KD-16270	8282.36
5	TP-21130	4454.06
6	AI-10855	4375.79
7	AB-10060	7755.63
8	RS-19420	3221.29
9	SB-20290	8057.89
10	GK-14620	6497.28
11	JS-15880	3868.01
12	CA-12775	3041.58
13	ES-14080	4657.93
14	AS-10285	3214.24
15	HK-14890	4877.80
16	PO-19180	4893.05
17	AB-10105	14473.57

Total rows: 294 Query complete 00:00:00.192